

# Raina Wu

✉ [rwu986@mit.edu](mailto:rwu986@mit.edu)    📍 Cambridge, MA    🔗 [sleepypandu.github.io](https://sleepypandu.github.io)    🌐 [sleepypandu](#)    📄 [raina-wu-890758326](#)

## Education

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>B.S. Massachusetts Institute of Technology</b> , Computer Science (Course 6-4, Artificial Intelligence and Decision Making) <ul style="list-style-type: none"><li>• GPA 5.0 / 5.0. Expected May 2027.</li><li>• <b>ML / AI:</b> 6.7960 Deep Learning, 6.7920 Reinforcement Learning: Foundations and Methods, 6.8300 Advances in Computer Vision, 6.7800 Inference and Information, 6.S184 Generative AI / Diffusion Lab (IAP); in progress: 6.8611 Natural Language Processing, 6.5940 TinyML and Efficient Deep Learning Computing, 6.4210 Robotic Manipulation</li><li>• <b>Theory and math:</b> 6.1220 Design and Analysis of Algorithms, 18.404 Theory of Computation, 18.701 Algebra I, 18.702 Algebra II, 18.510 Introduction to Mathematical Logic and Set Theory, 18.03 Differential Equations</li><li>• <b>Systems:</b> 6.1800 Computer Systems Engineering, 6.1910 Computation Structures, 6.1903 Introduction to Programming in C and Assembly, 6.1010 Fundamentals of Programming; in progress: 6.4400 Computer Graphics</li><li>• <b>Other:</b> 6.4590 Foundations of Information Policy, 24.09 Minds and Machines, 14.01 Principles of Microeconomics, 21H.152 Modern China, 24.02 Moral Problems and the Good Life</li></ul> | Cambridge, MA<br>Sept 2024 – May 2027 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|

## Experience

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| <b>IBM Research</b> , Research Intern — RL for LLM reasoning <ul style="list-style-type: none"><li>• Ran ablations and reproductions of existing self-distillation pipelines and uncovered failure modes of OPSD and SDPO.</li><li>• Ran a prefix-advantage study across Qwen3-1.7B/4B/8B characterizing where privileged-teacher signal localizes in a rollout. Submitted to the NeurIPS 2026 workshop on Transitioning from Pre-Training to Post-Training.</li><li>• Investigating other uses of distillation — whether privileged information can be used in a helpful, non-cheating way.</li></ul> | June 2026 – Aug 2026                  |
| <b>Hudson River Trading</b> , WITTI Wintern — quantitative finance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | New York, NY<br>Jan 2026 – Jan 2026   |
| <b>MIT-IBM Watson AI Lab</b> , Researcher — RL for generative models <ul style="list-style-type: none"><li>• Investigated RL fine-tuning of flow models for molecule and crystal generation.</li><li>• Co-author on <i>Tailored primitive initialization is the secret key to reinforcement learning</i> (ACL 2026); built the embedding-similarity pipeline used for the paper's diversity analysis, showing Tailor's warm-start data is more reasoning-diverse than the Standard-CoT and 4-STaR baselines.</li></ul>                                                                                 | Cambridge, MA<br>May 2025 – May 2026  |
| <b>Talkdesk</b> , Software Engineering Intern — multi-agent systems <ul style="list-style-type: none"><li>• Prototyped agentic user memory for a multi-agent system and a multi-agent evaluation framework for a production RAG pipeline.</li></ul>                                                                                                                                                                                                                                                                                                                                                    | June 2025 – Aug 2025                  |
| <b>MIT DeCoDE Lab</b> , Researcher — generative AI for design <ul style="list-style-type: none"><li>• Built the sketch-to-3D retrieval pipeline (2D sketch → nearest 3D car mesh) and benchmarked retrieval models and similarity metrics; co-author on the resulting ASME IDETC-CIE 2025 paper.</li></ul>                                                                                                                                                                                                                                                                                             | Cambridge, MA<br>Jan 2025 – Jan 2025  |
| <b>MIT Media Lab</b> , Researcher — AI security & agents <ul style="list-style-type: none"><li>• Built attestation for AI agents negotiating inside GPU-backed trusted execution environments; wrote circom circuits producing zkSNARK proofs that the negotiation ran as specified.</li><li>• Built tool-using agents (Gmail, Google Calendar, Shopify APIs) and an evaluation harness for multi-agent negotiation.</li></ul>                                                                                                                                                                         | Cambridge, MA<br>Sept 2024 – May 2025 |

- Proposed and evaluated an algorithm combining path-based link prediction with influence-maximization heuristics to identify future vital nodes in weighted collaboration networks.
- Introduced RA-2 and quasi-local RA-2, new link-prediction similarity metrics; built the simulation and evaluation pipeline (7 link-prediction metrics  $\times$  12 centrality measures  $\times$  8 selection algorithms, under simple and complex contagion); wrote the initial manuscript.
- Co-author (alphabetical order) on papers in *Physica A* (2026) and on arXiv (2024).

## Publications

---

### Tailored primitive initialization is the secret key to reinforcement learning

Aug 2026

Y. Yao, G. Zeng, R. Wu, Y. Zhang, D. Zhao, Z.-W. Hong, C. Gan

[doi.org/10.18653/v1/2026.acl-long.1537](https://doi.org/10.18653/v1/2026.acl-long.1537) (ACL 2026 (Volume 1, Long Papers), pp. 33300–33318)

### Influence prediction in collaboration networks: An empirical study on arXiv

2026

M. Lin, L. P. Schaposnik, R. Wu

[doi.org/10.1016/j.physa.2026.131451](https://doi.org/10.1016/j.physa.2026.131451) (Physica A: Statistical Mechanics and its Applications, 689, 131451)

### AI agents in engineering design: A multi-agent framework for aesthetic and aerodynamic car design

Aug 2025

M. Elrefaie, J. Qian, R. Wu, Q. Chen, A. Dai, F. Ahmed

[doi.org/10.1115/DETC2025-169682](https://doi.org/10.1115/DETC2025-169682) (ASME 2025 IDETC-CIE, Volume 3B, 51st Design Automation Conference)

### The social sphere model: Heuristic influence prediction in evolving networks

Feb 2024

M. Lin, L. P. Schaposnik, R. Wu

[arxiv.org/abs/2402.03522](https://arxiv.org/abs/2402.03522) (arXiv preprint arXiv:2402.03522)

## Honors

---

- **Putnam Mathematical Competition** — Top 220 (2025), Top 300 (2026)
- **High school:** Mathematical Olympiad Program (MOP) 2022, national top 60; USAJMO winner; USAMO qualifier; Math Prize for Girls national top 12 (2023)

## Teaching

---

- **Teaching Assistant, 18.404 Theory of Computation** — MIT, Fall 2026 (current)
- **Teaching Assistant, Athemath** — 2022–2024
- **Curriculum Developer, AlphaStar Academy** — 2021–2022

## Activities

---

- **MIT Syncopasian** — a cappella group ([syncopasian.com](https://syncopasian.com))

## Skills

---

### Programming

Python, C++, Java, SQL

### ML & Systems

RL for LLMs (PPO, GRPO, OPD, self-distillation), verl, vLLM, LoRA, FSDP, PyTorch, NumPy, multi-node GPU training (LSF), Docker, PostgreSQL, agentic systems, embedding calculations, circom / zkSNARKs

### Languages

English (native), Mandarin Chinese (fluent)